

Prayas 2.0 2027 (OG)

Physical Chemistry

Some Basic Concepts of Chemistry

DPP: 5

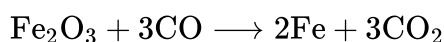
Time Duration - 30 min

- Q1** Find the mass of 50% CaCO_3 which will produce 136 g of CaSO_4 ? (Molar mass of $\text{CaCO}_3 = 100$ g; Molar mass of $\text{CaSO}_4 = 136$ g)
 $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$
 (A) 100 g
 (B) 200 g
 (C) 300 g
 (D) 400 g

- Q2** If 300 g of CaCO_3 on reaction with excess of H_2SO_4 gives 272 g of CaSO_4 . Find the % age yield of product?
 (Gram atomic mass of C = 12g, H = 1g, O = 16g, S = 32g and Ca = 40g)
 $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
 (A) 66.66% (B) 33.33%
 (C) 50% (D) 25%

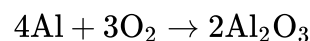
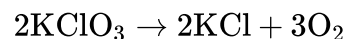
- Q3** For the reaction: $2\text{X} + 3\text{Y} \rightarrow 3\text{Z}$, the combination of 2.00 moles of X with 2.00 moles of Y produces 1.75 moles of Z. What is the percent yield of this reaction?
 (A) 43.8% (B) 58.3%
 (C) 66.7% (D) 87.5%

- Q4** Fe_2O_3 reacts with excess CO at a high temperature according to the equation below:



If 6.50 g of Fe_2O_3 yields 3.85 g of Fe, what is the percentage yield of the reaction?
 (A) 59.2% (B) 69.9%
 (C) 76.3% (D) 84.7%

- Q5** Based on the following consecutive reactions



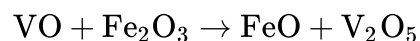
2 moles of KClO_3 of 50% purity on strong heating and then with excess of Al, Al_2O_3 formed is

- (A) 2 mol
 (B) 1 mol
 (C) 3 mol
 (D) 1.5 mol

- Q6** The weight of lime obtained by heating 200 kg of 95% pure lime stone is:
 (A) 98.4 kg
 (B) 106.4 kg
 (C) 112.8 kg
 (D) 122.6 kg

- Q7** For the reaction $2\text{P} + \text{Q} \rightarrow \text{R}$, 8 mol of P and 5 mol of Q will produce
 (A) 8 mol of R
 (B) 5 mol of R
 (C) 4 mol of R
 (D) 13 mol of R

- Q8** 5 mol of VO and 6 mol of Fe_2O_3 are allowed to react completely according to the reaction



The number of moles of V_2O_5 formed is :

- (A) 6 (B) 2
 (C) 3 (D) 5

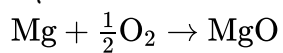
- Q9** Find the volume contraction if 10 L of Butane (C_4H_{10}) under combustion with sufficient amount of oxygen?
 (A) 35 L



- (B) 25 L
- (C) 15 L
- (D) 3.5 L

Q10 1.2 g of Mg (At mass 24) will produce MgO

equal to



- (A) 0.05 mol
- (B) 40 g
- (C) 40 mg
- (D) 4 g



Answer Key

Q1 (B)

Q2 (A)

Q3 (D)

Q4 (D)

Q5 (B)

Q6 (B)

Q7 (C)

Q8 (B)

Q9 (A)

Q10 (A)



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